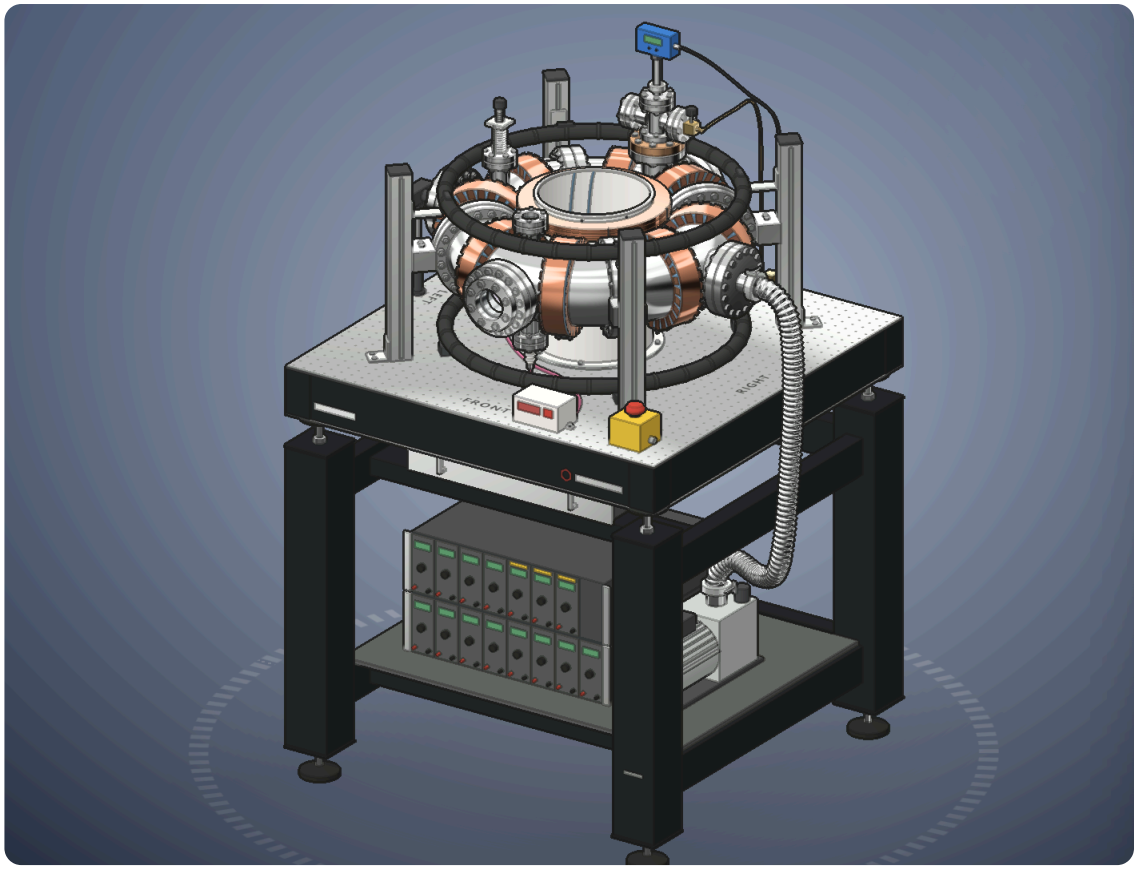
User guide

SPM1 anatomy

How to use the interactive 3D anatomy of the MinimaK SPM1, a tabletop magnetised torus by Rideo Systems



What it is

A 3D model you can turn, take apart and read. For each of the machine's 29 components it shows what the part is, how it attaches to the next one, what it is made of, and where each number comes from. It does not simulate the plasma physics.

Opening it

Open `index.html` in a recent browser (Chrome, Edge, Firefox or Safari). There is nothing to install. The first time it opens, it needs an internet connection to load its 3D library.

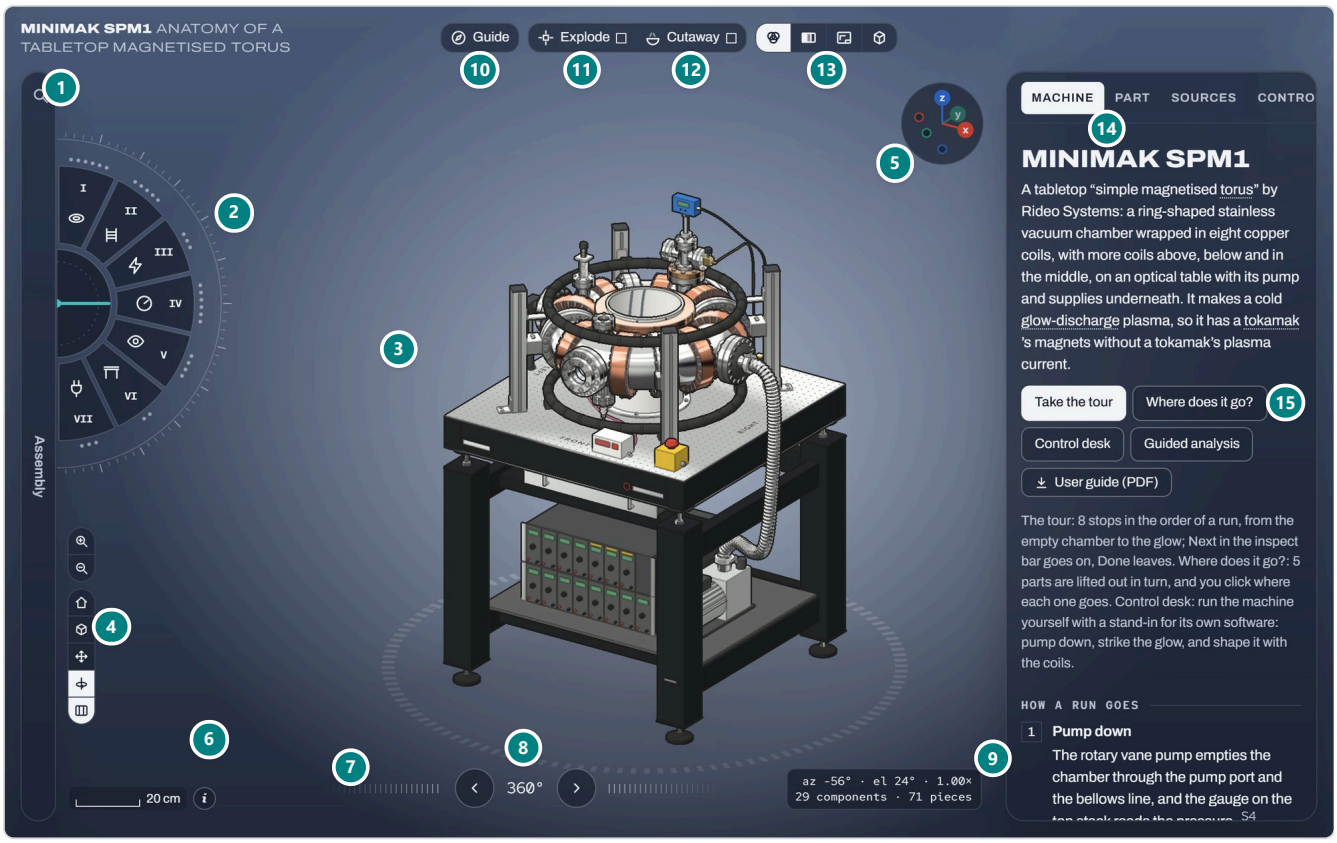
This guide is on the page too: **User guide (PDF)** in the Machine tab downloads it.

In this guide

1 The page at a glance 2
2 Guided analysis 3
3 Moving round the model 4
4 Finding a part 5
5 Inspecting a part 6
6 Taking a part apart 7
7 Sheets and Show me 8
8 Where the numbers come from 9
9 The whole machine: Explode and Cutaway 10
10 Learning the machine: the tour and the quiz 11
11 Running it: the control desk 12
12 Four themes 13
13 Phones and tablets 14
14 Links, settings and access 15
15 Keys and gestures 16

1 The page at a glance

The machine stands in the middle of the window, the inspector on the right. On a first visit the assembly list is folded to a thin rail on the left, with the groups' wheel beside it, so the model has the room. The bar at the top changes the whole machine; the view's own controls stand round its edges.

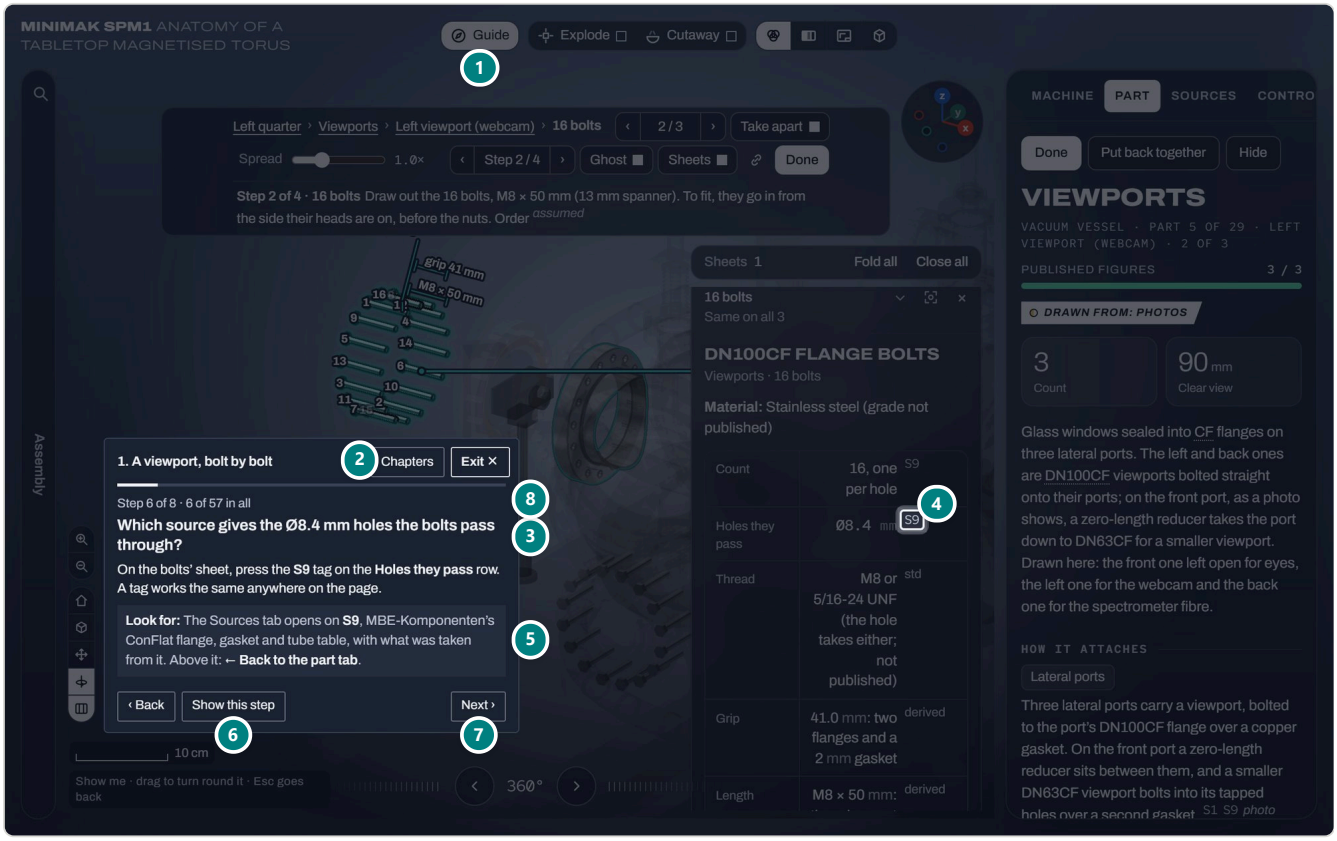


1. **Assembly list.** All 29 components in seven groups, with a search box. Click its magnifier, or press `/`, to open it with the cursor in the search (page 5).
2. **Groups' wheel.** The seven groups on a half-wheel while the list is folded. Rest the pointer on one and its parts fan out as pictures (page 5).
3. **The 3D view.** Drag to turn, right-drag to pan, scroll to zoom. Until you touch it, it turns slowly on its own.
4. **View buttons.** Zoom in and out, the home view, **Views** (look from a side), pan mode, the axis lock and the side panels (page 4).
5. **Axis triad.** Shows which way is up; click an axis end to look along it.
6. **Scale bar.** True at every zoom: the view has no perspective, so sizes can be compared anywhere in it.
7. **Hint.** What a click or a drag does right now. It changes as you go.
8. **Turntable.** ‹ and › turn the machine an eighth of a turn.
9. **Readout.** The view's angles and zoom, and what is shown or selected.
10. **Guide** opens the guided analysis: the viewer taught on the SPM1, one question at a time (page 3).
11. **Explode.** Takes every component off the machine at once (page 10).
12. **Cutaway.** Cuts the top off the vessel to show the plasma inside.
13. **Theme.** Colour, Greyscale, Blueprint or Hologram (page 13).
14. **Inspector.** Four tabs: **Machine** (what it is, how a run goes), **Part** (the selected part), **Sources** and **Controls**.
15. **Take the tour** and **Where does it go?**, a guided tour and a quiz (page 11); under them, **Control desk** (page 12), **Guided analysis** and **User guide (PDF)**, which downloads this guide.

> Make room. Drag the edge between a side panel and the view to size the panel; double-click the edge for the default width. The panels button at the view's left edge, or `F`, folds both side panels so the model gets the whole window.

2 Guided analysis

New to the page? **Guided analysis**, the first button in the top bar (**Guide** on a narrow window), teaches the viewer on the SPM1 itself, one question at a time: where a part is, how it attaches, and where its numbers come from. You use the page's own controls: the guide darkens the page round the one it is showing, and only that control takes a click.



1. **Guide** opens it, and closes it again. The Machine tab has the same button.
2. **Chapters** jumps to any of the seven. **Exit** leaves, and gives back your view, sheets and settings.
3. **The question** the step asks, then the real control to use, by its name on the page, with its key.
4. **The spotlight**: the page darkens round that control, ringed; use the control itself. When it is hidden (under ⋯ on a phone, or in a folded panel), the spotlight shows the way in.
5. **Look for**: what shows when it works. It turns into a tick, *That's it*, once the page is in that state.
6. **Show this step** does it for you, with the same action a click makes.
7. **Next** › and ‹ **Back** move between steps; the last one is **Finish**.
8. **Where you are**: the step in its chapter, and in all 57.

Seven chapters

1. **A viewport, bolt by bolt**: find it, take it apart, measure one bolt, follow its number to its source and back, and watch it go home.
2. **Orient the view**: turn and the turntable, pan, the axis lock, zoom, the scale, a side, home, the panels.
3. **Find hardware**: the groups' wheel, search for M8, select or inspect, hide and show, what is listed but not drawn.
4. **See inside**: Cutaway, Explode, Spread, the themes, Blueprint's dimensions.
5. **Inspect a connection**: pieces, Ghost, How it attaches, steps, a joint opened, Turn to see.
6. **Read the evidence**: callouts, previews, sheets, Show me, sources, tags, the glossary.
7. **Use the result**: Copy link, Back, the run, the tour, the quiz, the control desk, settings.

Good to know

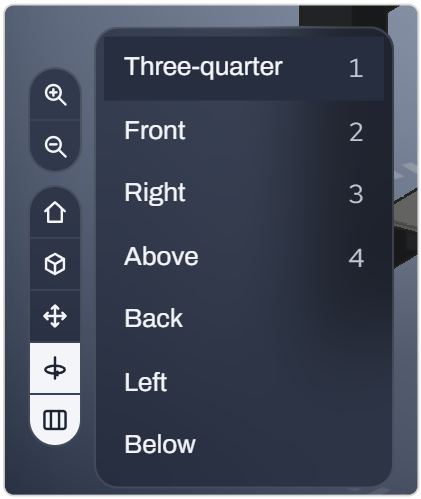
- Only the spotlight and the card take a click, and only the keys the step names work. Once a step is done, the shade lightens to show what changed.
- Nothing is saved, and nothing is added to the browser's history. **Exit** or **Finish** puts back your view, the part and sheets you had open, hidden parts, panels, theme and settings.
- **Take the tour** and **Where does it go?** close the guide first. Open it again and **Resume** picks up at the step you left.
- From the keyboard, focus goes to the question, and Tab reaches **Show this step** and **Next** ›. Each step is announced to screen readers.
- The card sits at the view's lower left, or under the view on a phone. With reduced motion, every change happens at once.

3 Moving round the model

The model turns like a turntable: sideways spins it about its own axis, the middle of the ring on the floor, and up and down tilts it. The vertical always stays up, so you never lose your bearings.

To	Mouse	Touch	Keys
Turn	Drag; or ‹ › on the turntable under the machine, an eighth at a time	One finger, or the turntable	Arrow keys
Pan	Right-drag, Shift-drag or middle-drag; or switch on the pan button, then drag	Two fingers	Shift + arrows
Turn about the machine or the view	The axis button. Lit, as every visit starts, the machine turns in place wherever you pan it; off, a turn goes about the middle of the view	The same	R
Zoom	Scroll wheel (it zooms toward the pointer), or the zoom buttons	Pinch	+ −
Home view	The home button, or double-click empty space	The home button	0
Fit everything shown	—	—	A
Look from a side	**Views** menu, or click an axis end on the triad	The same	1 three-quarter, 2 front, 3 right, 4 above
Fold the side panels	The panels button	The panels button	F

Keys work once the model has focus: click it first. While inspecting a part, the home view frames that part instead of the whole machine, and a turn always goes about the part.



The view buttons and the **Views** menu. From the top: zoom in, zoom out, home, Views, pan mode, the axis lock (lit), side panels. The menu works from the keyboard too: ↑ ↓ move, Enter picks, Esc closes.



The axis triad

The filled ends are the positive axes, with their letters (x red, y green, z blue, z up); the hollow rings are the negative ends. Click any end to look from that side.

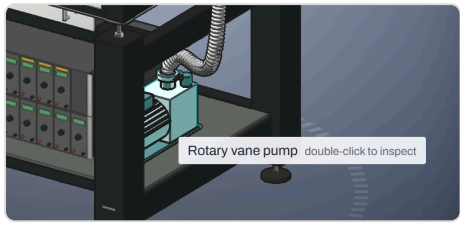
The table's sides

The table's edges are printed with the quarters above them: **Front**, **Right**, **Back** and **Left**. Parts are named by these sides, as in *Left viewport* or *Right–back joint*.

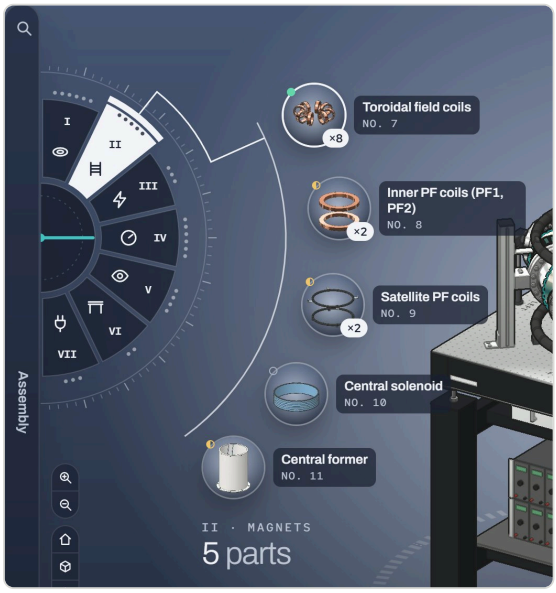
> **Lost?** Press 0 or double-click empty space to go back to the home view.

4 Finding a part

Point at any part to see its name; click it to select it. A selected part gets a teal layer and outline, the rest of the machine fades back, and the **Part** tab tells you about it, with **Inspect on its own**, **Zoom to it** and **Hide**.



Pointing at a part names it and lights it.

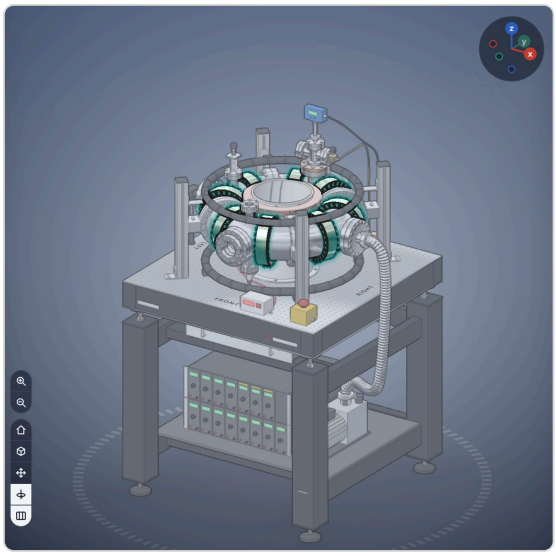


The magnets' five parts, fanned out from the wheel.

The groups' wheel

While the assembly list is folded, the seven groups stand on a half-wheel at the view's left edge: a wedge each, with its sign, its numeral and a dot for each of its parts. The group of the selected part is outlined.

- **Rest the pointer** on a wedge, or click it: the group's parts fan out beside the wheel as pictures, each with its count and its basis dot.
- **Click** a picture to select the part; **double-click** to inspect it. Pointing at one lights it on the model.
- Moving away folds the fan; one opened with a click stays until Esc or a click elsewhere. From the keyboard, Enter opens a group and the arrows step through its parts.
- On a phone, or in a window too short for it, the wheel steps aside and the list is the way in.



The toroidal field coils selected, from the model, the wheel or the list.

The assembly list

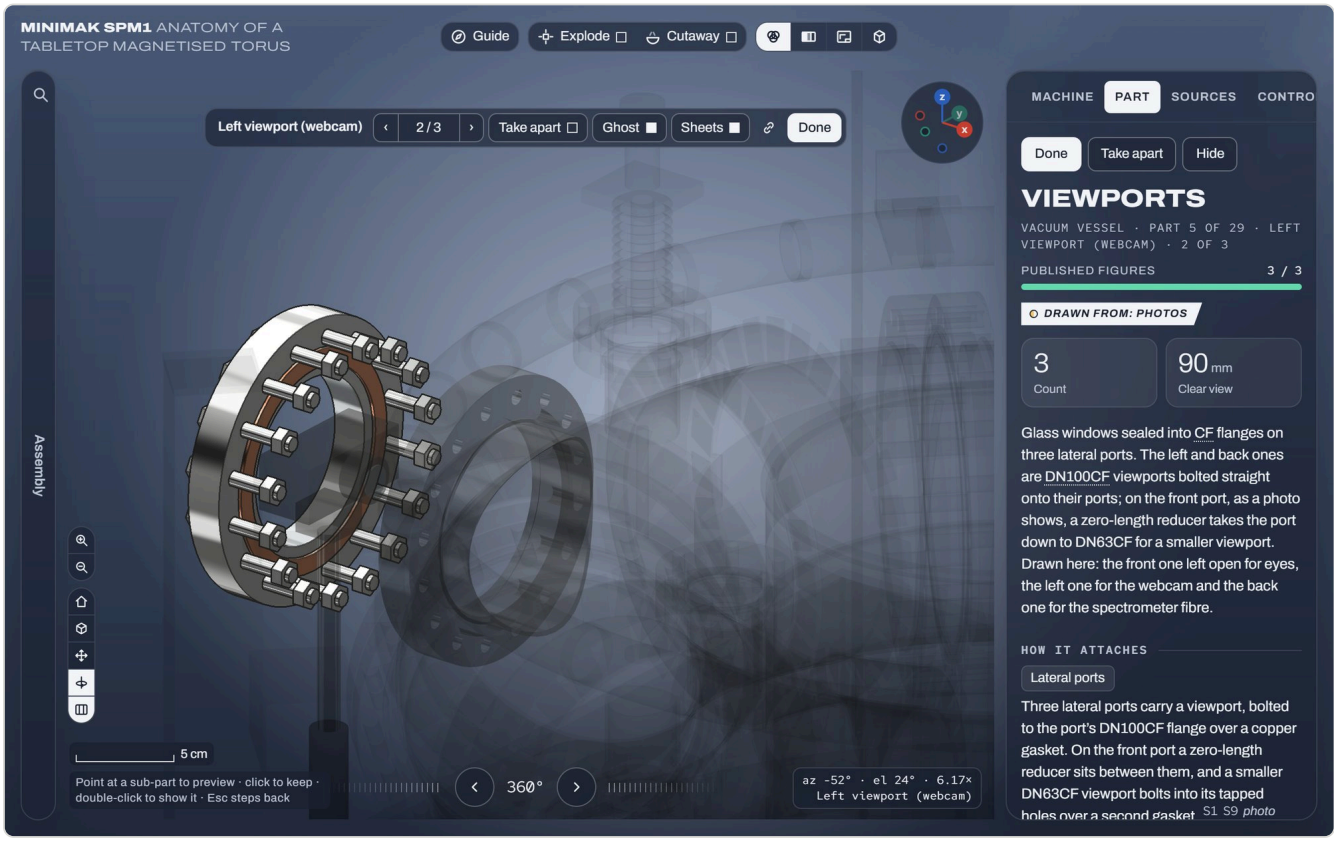
- **Click** a name to select the part; **double-click** to inspect it.
- **×4** after a name is how many pieces the part has.
- **The eye** hides a part, or a whole group, and shows it again.
- **The dot** is the part's basis: ● spec, ◐ photos, ○ assumed.

Search

The box at the top of the list searches names, specs, descriptions and every sub-part's sheet. Try a size (*M8* lists every M8 bolt and nut), a seal (*o-ring*), or a question in plain words: *the gasket between the right and back quarters*. The best matches are listed, with matching pieces under their part; click one to inspect it. A line under the box says which words found nothing.

5 Inspecting a part

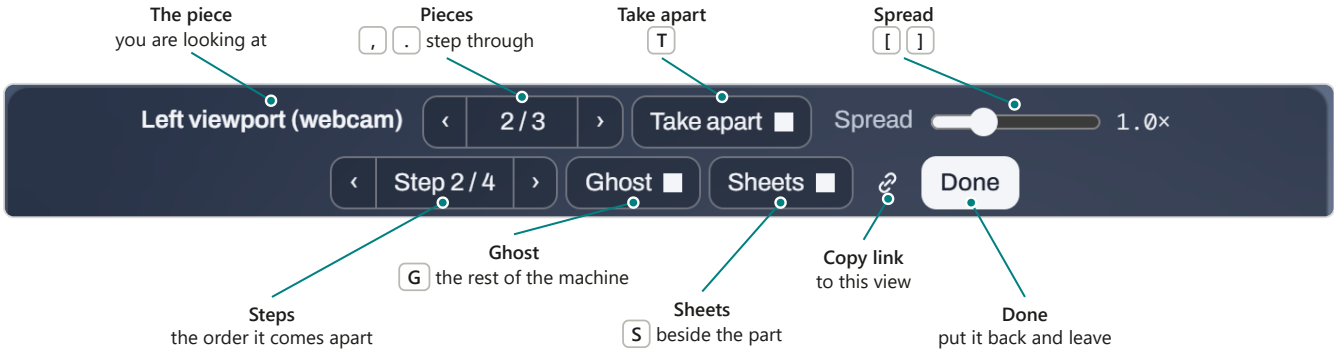
Inspecting shows one component on its own. The rest of the machine fades to an x-ray ghost, and the part lifts out while the camera glides round to frame it, so you can see where it comes from.



Inspecting the viewports, on the left one (piece 2 of 3). The parts it bolts to are drawn half solid with their bolts and nuts; the rest of the machine is a ghost.

To inspect a part: double-click it in the view or in the list, press **Inspect on its own** in the Part tab, or select it and press I.

The inspect bar



The Part tab

- **What it does**, and what it is made of.
- **How it attaches**: one sentence for each part it connects to. Click a name to inspect that part; ← **name** in the bar, or Backspace, goes back.
- **Made of**: its sub-parts, numbered. Each has a sheet (page 8).
- **Pieces**: where each piece of a repeated part sits, and what tells them apart.
- **Specification**: every row carries a source tag (page 9).

Getting back out

Esc steps back one level at a time: the preview, Show me, the opened joint, the step, the single piece, the part, then the selection. Kept sheets stay.

Done reverses the whole inspect at an explode's pace: the part goes back together, flies home while the camera returns to your earlier view, and the machine turns solid again.

A part with several pieces opens on all of them; click one to pick it, or use the arrows beside **All**.

6 Taking a part apart

Take apart (T) separates the inspected part into its sub-parts, each along the way it really comes off: bolts slide out, nuts come off the back, gaskets come out from between their flanges, a KF clamp opens. A weld never comes apart.



1. **Star order.** When a step takes off a ring of bolts or nuts, each head is numbered with its turn in the star (cross) tightening pattern: 1, 11, 6, 16 … round a 20-bolt flange. Point at one bolt to name it: *Bolt 7 of 20 · star 3*.
2. **How to do the step:** the size, length and spanner (*Draw out the 16 bolts, M8 × 50 mm, 13 mm spanner*), which way a knife edge faces, and that a copper gasket is never reused.
3. **Steps.** ‹ **Step 2 / 4** › walks through the order the sub-parts come off: nuts, then bolts, then the flange, a gasket last. **All** has every sub-part off; fitting runs back up the steps. No procedure is published, so the order is tagged assumed.

More

- **Spread** (slider, or [ ]) sets how far apart the sub-parts go.
- A **dotted trail** from a hollow ring shows where the sub-part under the pointer sat, and the way it came off.
- **Turn to see** appears when the sub-parts come apart toward you and hide each other. It turns the view so they lie side by side. The view never turns by itself.

Open this joint

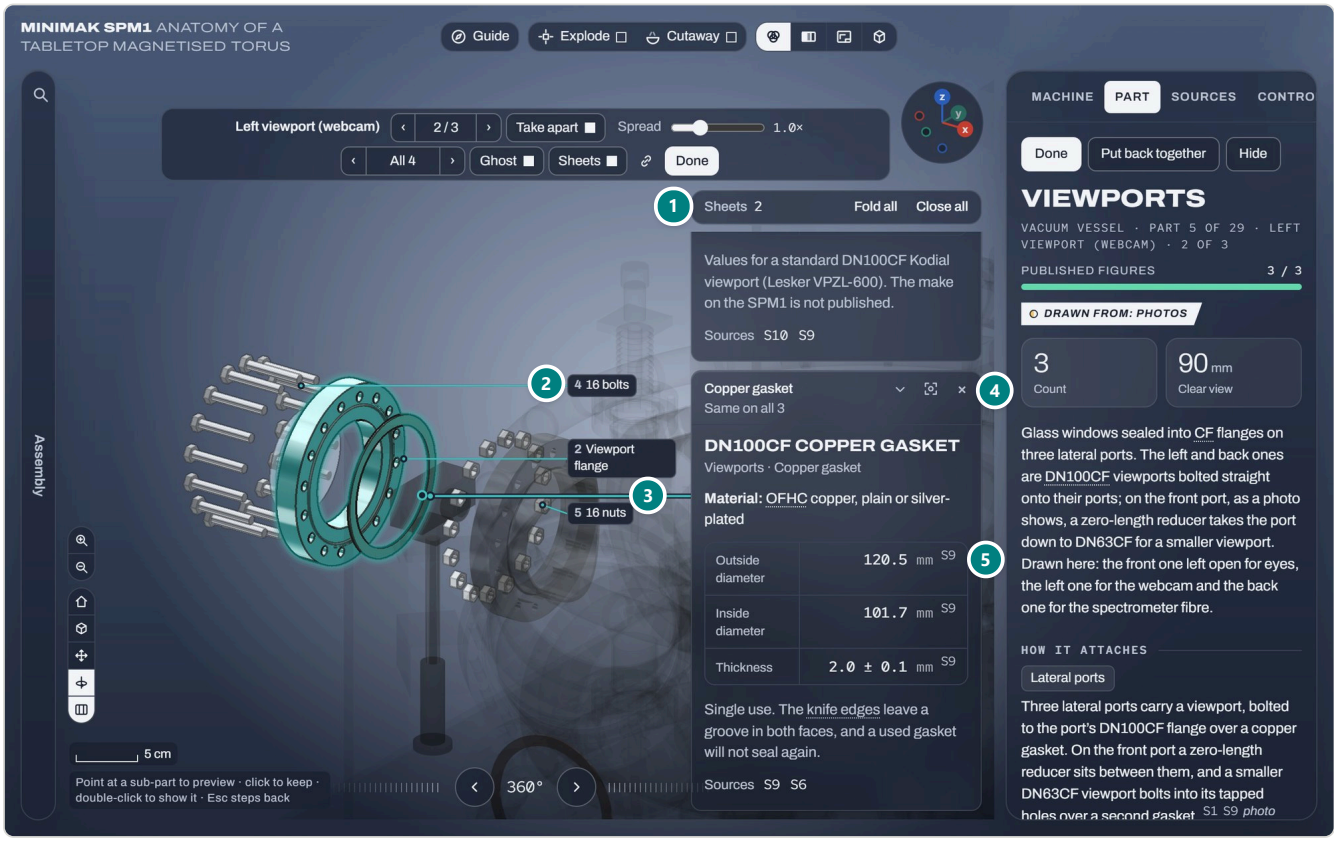
On a copper gasket or a joint flange, **Open this joint** (J) opens the whole vessel joint: the 20 nuts off in star order, the 20 bolts out, the two quarters eased apart, then the gasket out. Press it again, or Esc, to close it.



The right–back joint opened at step 2: its 20 bolts drawn out, each numbered with its turn in the star order.

7 Sheets and Show me

While you inspect, every sub-part has a technical sheet: its material, dimensions and notes, with a source on every row. Sheets sit in a column on the right of the view, where they never cover the part.



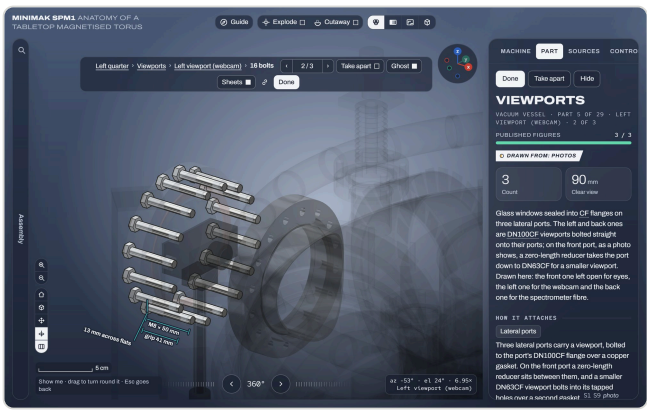
1. **The column** counts the kept sheets. **Fold all** folds them to their bars; **Close all** closes them.
2. **Callouts** name the sub-parts beside the part, numbered as the Made of rows. Click one to preview its sheet.
3. **The leader line** joins a sheet to its sub-part.
4. **A sheet's bar:** the chevron folds it, the frame button is Show me, × closes it. *Same on all 3*: a sheet belongs to the sub-part, so it is the same on every piece.
5. **Source tags** on every row say where the number comes from (page 9).

Preview and keep

- **Preview:** rest the pointer on a sub-part. Its sheet opens with a dashed edge, labelled *Preview*, and stays until you rest on another one.
- **Keep:** click the sub-part, press **Keep** on the preview, or click its row under **Made of**.
- **Float:** drag a sheet by its bar to anywhere on the view. **Tidy** (L) puts floating sheets back in the column.
- Each part remembers its sheets, folds and places for when you come back.

Show me

Brings one sub-part close up and dims the rest of the part. Use the frame button on a sheet's bar or a Made of row, Z, or double-click the sub-part. The bar shows where it is (*Left quarter › Viewports › Left viewport › 16 bolts*), and a ring of bolts gets its sizes drawn on. Esc goes back.



8 Where the numbers come from

Every number on the page carries a tag that says where it comes from, so you can tell a published figure from one read off a photo or picked for the drawing.

Tag	Meaning
S1 … S11	Published in that source. Click the tag to open it in the **Sources** tab; ← **back** returns you.
photo	Scaled from the photos in the 2025 webinar on the Minimak (S4).
measured	Measured on a top-down photo of an SPM1, with the camera fitted to parts of known size.
std	Follows from the ConFlat standard (S9); the SPM1's own part is not published.
derived	Worked out from tagged numbers: the vessel volume, the field on the axis.
assumed	Not published; picked for the drawing.

A key at the foot of each sheet and of the Part tab explains the tags it uses.

How far to trust a shape

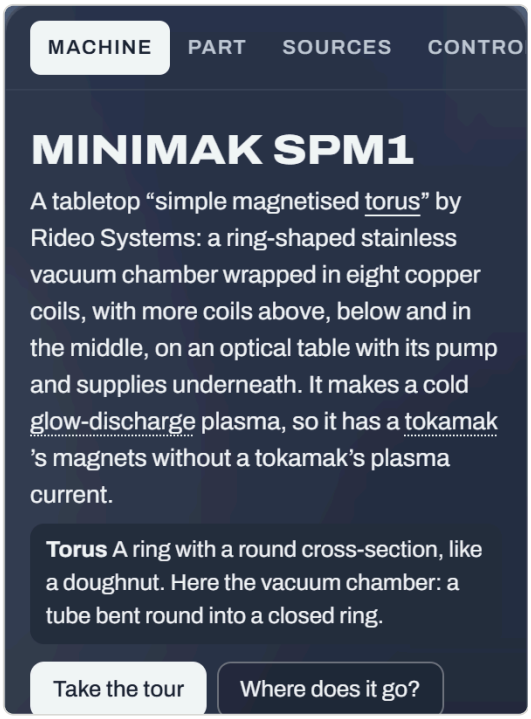
The Part tab says what the part's shape and position are drawn from, with the same dot as the list:

- ● **published spec**: the size or placement is published.
- ◐ **photos**: scaled or placed from photos.
- ○ **a guess**: placed where it is plausible.

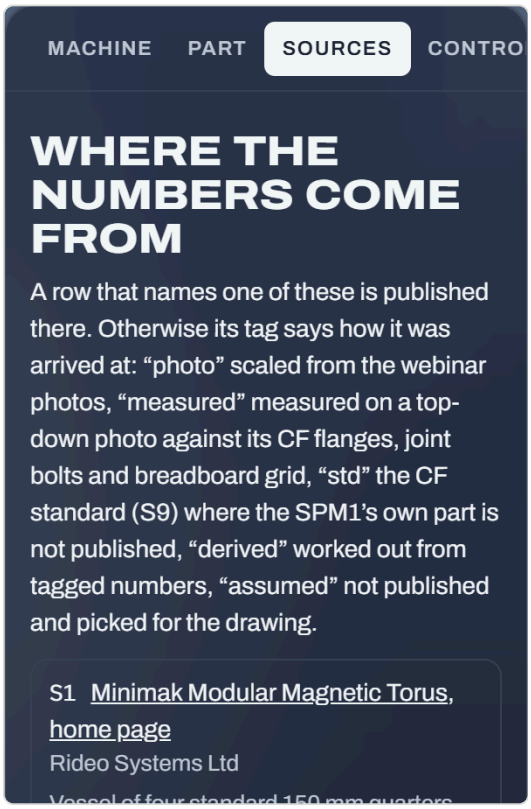
This is about the drawn shape, not the numbers in the rows, which have their own tags.

Words explained where they are used

About two dozen terms a first-year student may not know (torus, poloidal, ConFlat, knife edge, Pirani gauge, mbar and more) have a dotted underline the first time they appear. Click one, or Tab onto it, to open its definition under the paragraph; click again to close it.



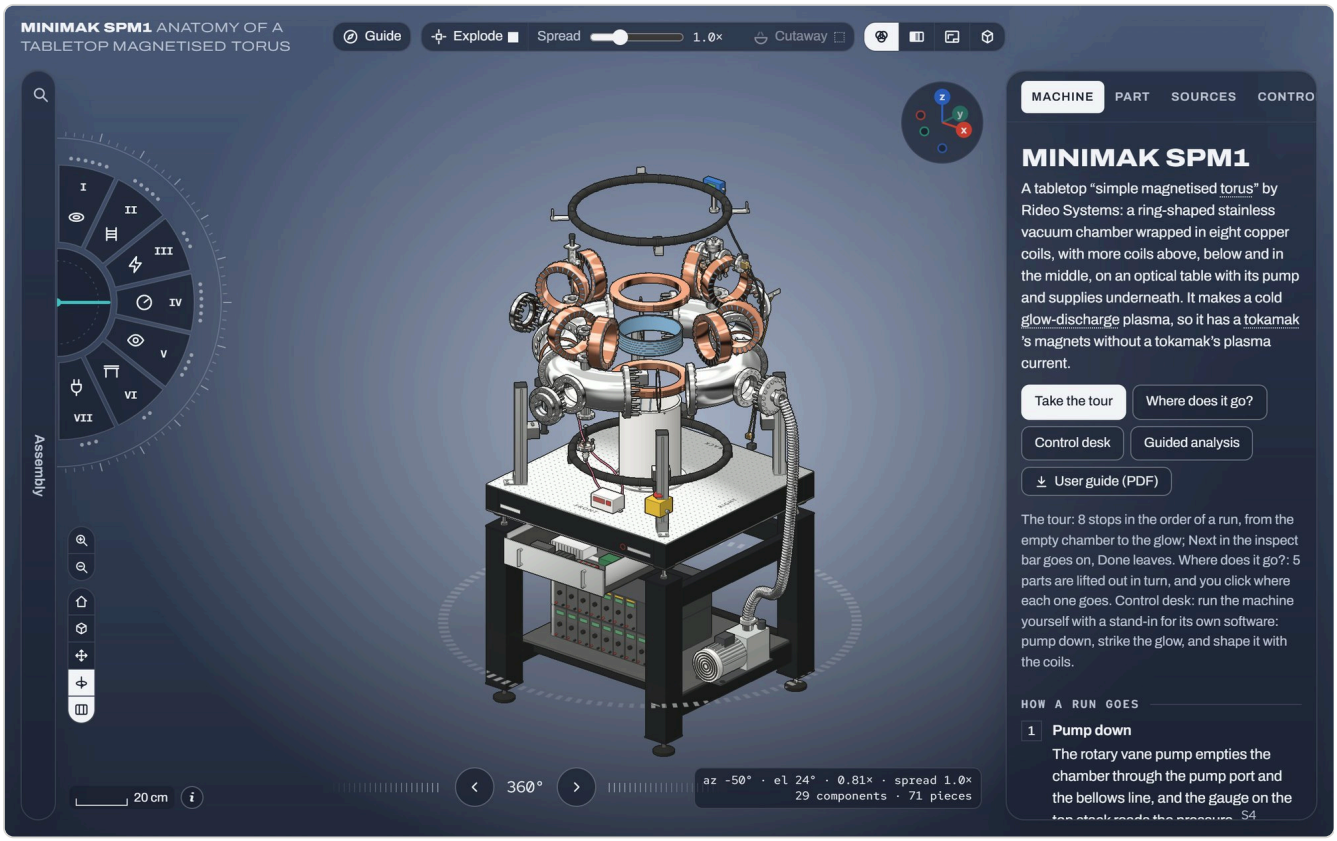
A click on *torus* opens its definition.



The Sources tab: every reference, what was taken from it, and how many rows cite it.

9 The whole machine: Explode and Cutaway

Explode (E) takes every component off the machine at once, so you can see how it all fits together.



Exploded and zoomed to fit (A). The quarters move out, the coils and ports lift off them, the PF and satellite coils rise, and the table, supplies and pump drop away. Cables and the pump hose stay plugged in at both ends.

Explode

- **Spread**, the slider beside the switch, or [ ], sets how far apart the parts go (0.4× to 2.5×).
- The plasma is off while the machine is apart. The windows are clear glass, so you can look into the vessel through any of them.

Cutaway and hiding

- **Cutaway** (C) cuts away the top of the vessel to show the glow inside. Every visit opens with it off, the vessel closed.
- **Hide** a part with its eye in the list, the **Hide** button in the Part tab, or H for the selected part. The eye on a group hides the whole group.
- **Fit** (A) zooms to whatever is shown, at the current angle.

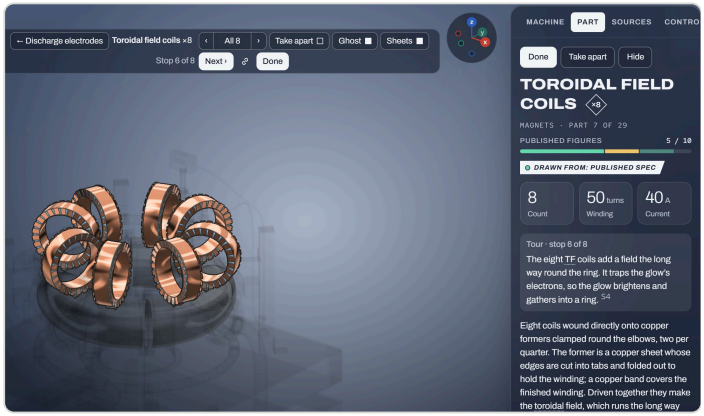
10 Learning the machine: the tour and the quiz

The **Machine** tab explains what the SPM1 is and how a run goes, in four steps: **pump down**, **let in a little gas**, **strike the glow** and **add the magnetic field**. Each step names the parts it uses; click a name to select that part. The order follows the 2025 webinar on the Minimak (S4); no operating procedure is published. Beside the tour's button, **Control desk** lets you run the machine yourself (page 12), and **Guided analysis** teaches the viewer itself (page 3).

Take the tour

Inspects the parts in the order of a run, in eight stops from the empty chamber to the glow: the vessel, the pump, the gauge, the gas valve, the electrodes, the TF coils, PF1 and PF2, and the plasma.

- At each stop, a line in the Part tab says why that part matters at that point in the run.
- **Next** › in the inspect bar goes on; it shows *Stop 6 of 8*.
- You can look around at any stop: take the part apart, open sheets, follow links. Only **Done** leaves the tour.

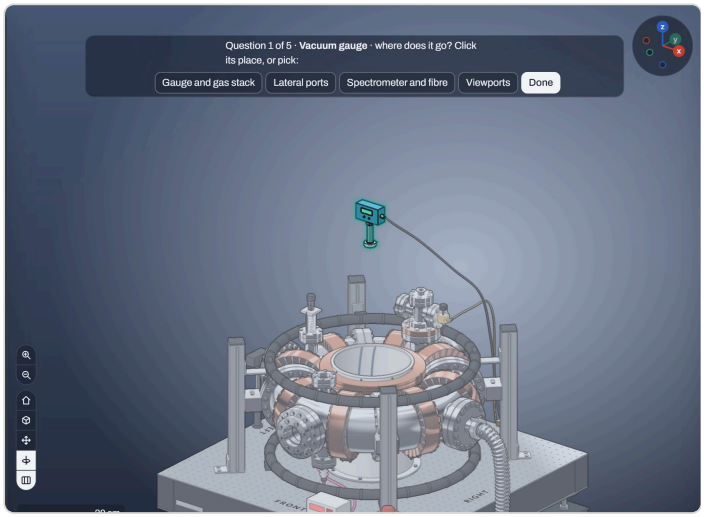


Stop 6 of 8: the toroidal field coils. The Part tab says why they matter now.

Where does it go?

A quiz of five questions. Each lifts a part out above the machine; click the place it belongs on the model, or pick one of four parts in the bar.

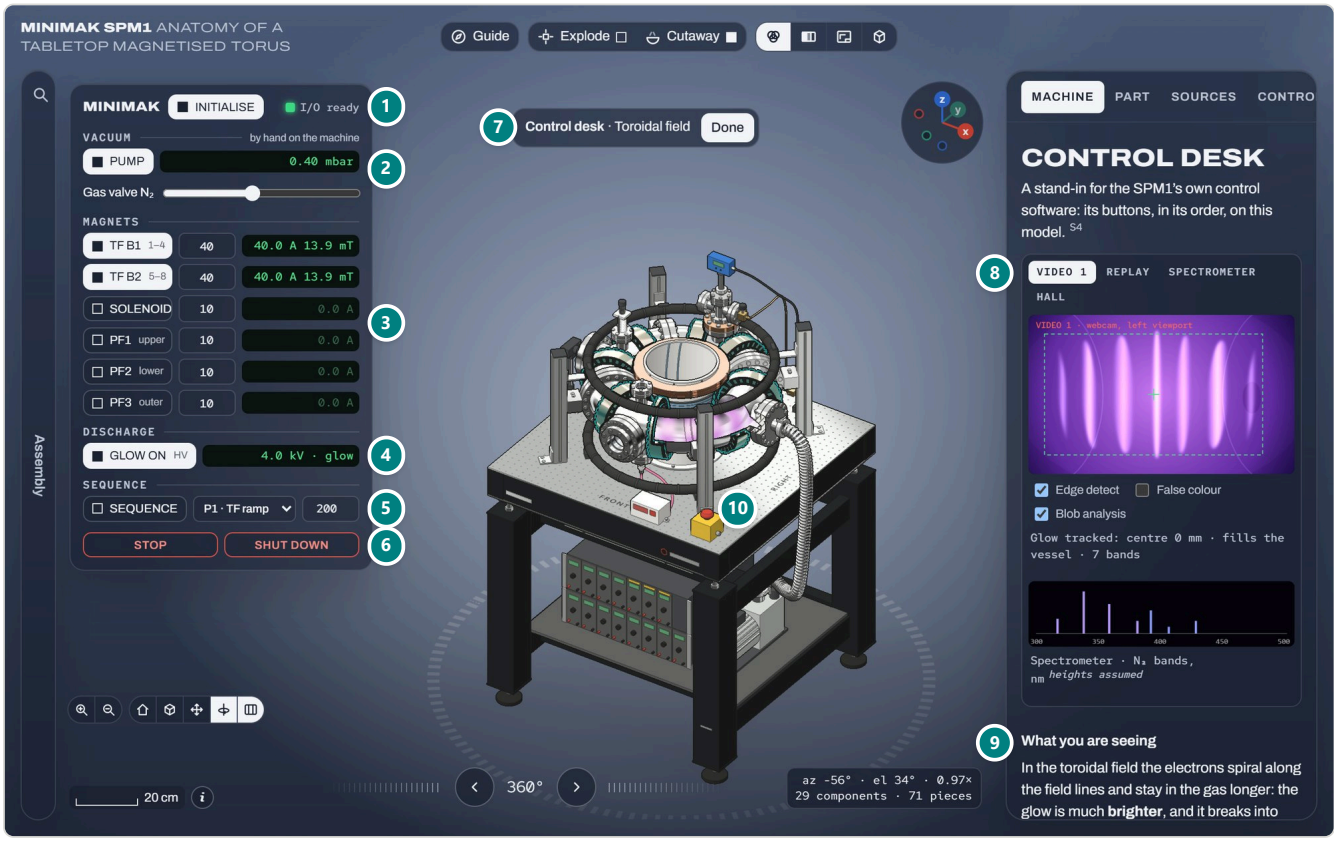
- A wrong answer names what you clicked and gives you one more try. Then the part flies home, its place is outlined, and the bar says how the two are joined.
- The round is scored on first tries; **New round** asks five more.
- A lecturer can send a round as a link (page 15).



The vacuum gauge is lifted out: click where it belongs, or pick in the bar.

11 Running it: the control desk

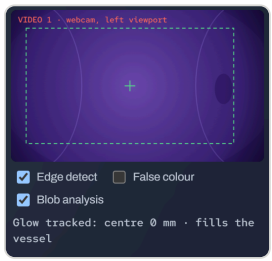
Control desk, in the Machine tab, lets you run the machine with a stand-in for its own control software: the real program's buttons, in its order. The glow on the model answers as it does in the maker's demonstrations.



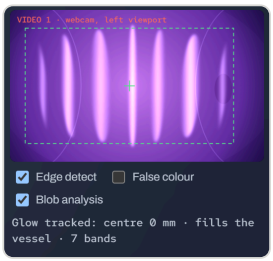
1. **INITIALISE** opens the link to the supplies; the light beside it turns green. No supply switches on before it.
2. **Vacuum**: the **PUMP**, the **gas valve** and the gauge's pressure. On the real machine these are worked by hand.
3. **Magnets**: the two TF banks, the solenoid and PF1 to PF3. Each has a switch, a setpoint up to 40 A, and the current it has reached. The coils that are on are lit on the model.
4. **GLOW ON**: the high voltage. Its readback says whether the glow has struck.
5. **SEQUENCE** runs a program, a step each interval: a TF ramp, a PF sweep or pulses.
6. **STOP** switches the high voltage and the coils off; **SHUT DOWN** the pump, the gas and the link too.
7. **The bar** says what state the machine is in. **Done**, or Esc, leaves and puts your view back.
8. **The monitor**: **VIDEO 1**, the webcam's picture with the tracking's box and centre; **REPLAY**, the last sequence's frames; **SPECTROMETER**; **HALL**, the fields.
9. **What you are seeing**, and why, with its source. Under it, five things to try, ticked as you do them.
10. **The emergency stop.** Click it on the table: the high voltage and every coil drop at once. INITIALISE resets the desk.

A run, step by step

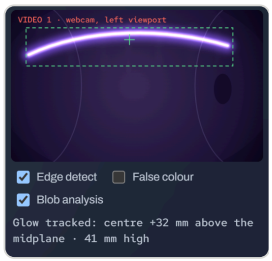
1. **INITIALISE**, then **PUMP**: the pressure falls.
2. Open the **gas valve** until the gauge reads between 0.1 and 2 mbar.
3. **GLOW ON**: an even violet fills the vessel.
4. Switch the **TF** banks on and raise their setpoints: the glow brightens and breaks into bands.
5. Switch **PF1** and **PF2** on with different currents: the glow gathers into a ring, and moves up or down.



No field: an even glow.



Toroidal field: brighter, in bands.

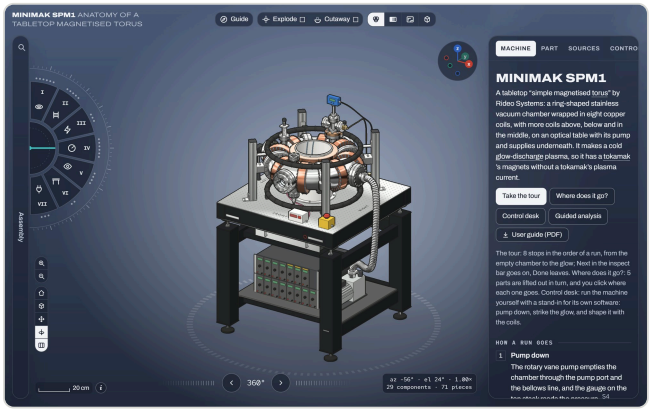


PF1 above PF2: a ring, moved up.

A sketch, not a simulation. The glow follows a few fixed rules taken from the maker's demonstrations. The page says which of its numbers are assumed, and time is compressed: the real pump takes minutes.

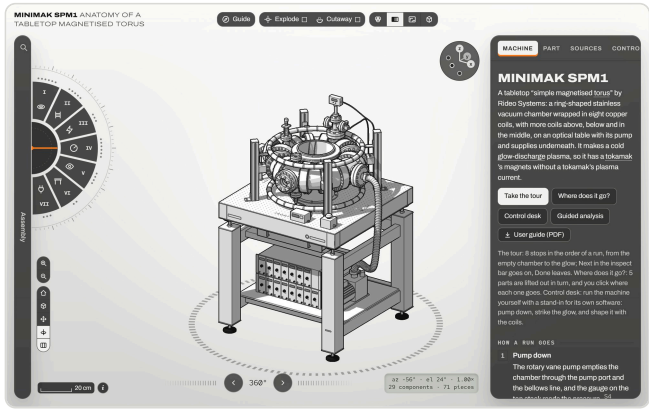
12 Four themes

The last four buttons in the top bar switch between four ways of drawing the same machine. Everything works the same in every theme, and the page remembers your choice. The page's name, at the top left, is set the same way in all four: wide capitals, the name heavy and what this is light.



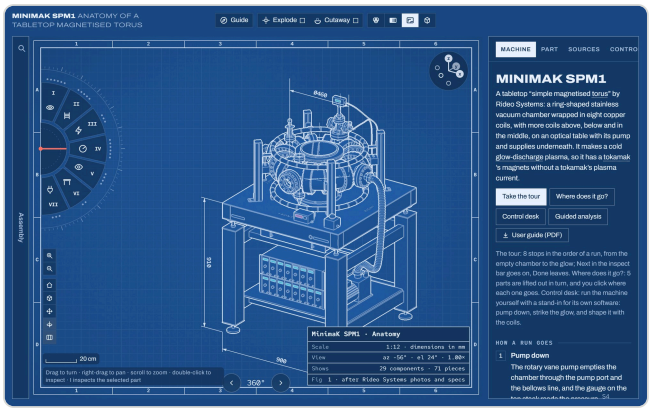
Colour

Solid shaded parts in their own colours with inked outlines, on a slate-blue stage. The bare stainless and copper reflect a small studio, polished as on the real machine. The selection is teal.



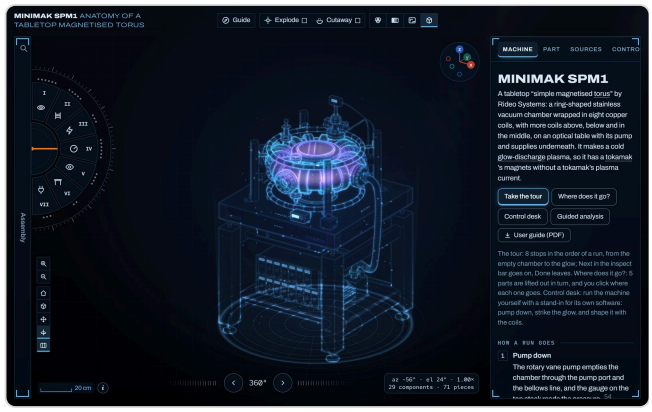
Greyscale

A white, CAD-style rendering with five flat shades and hard shadows, in a white studio. Only the plasma (pink-violet) and the selection (amber) are in colour.



Blueprint

White drafting lines on blue paper, set out as a drawing sheet: a frame with zones, centre lines, a grid, and a title block with the scale. Three dimensions (910, 900, Ø460 mm) can be dragged further off or along; double-click one to put it back. The selection is a red pencil mark.



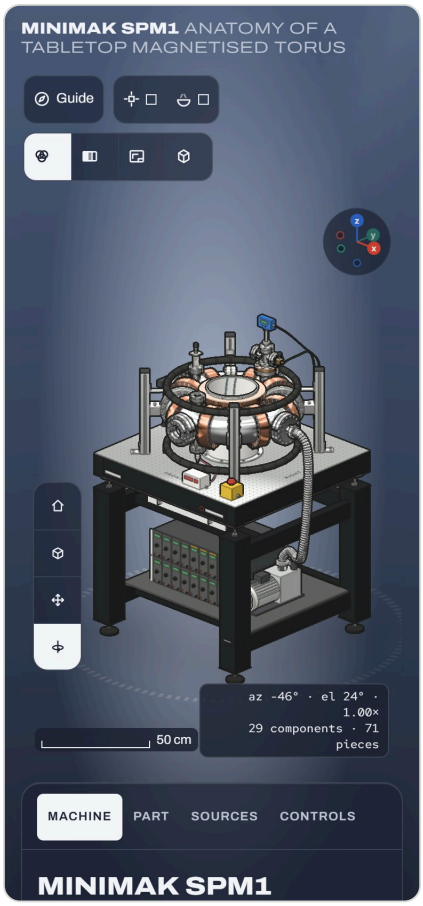
Hologram

Ice-blue light on near-black, from a projector under the stand: the torus is lit most and the inside of the machine shows through, with the panels and sheets as an ice-blue HUD. The plasma glows violet and the selection orange. With no theme chosen, a dark system setting opens this one.

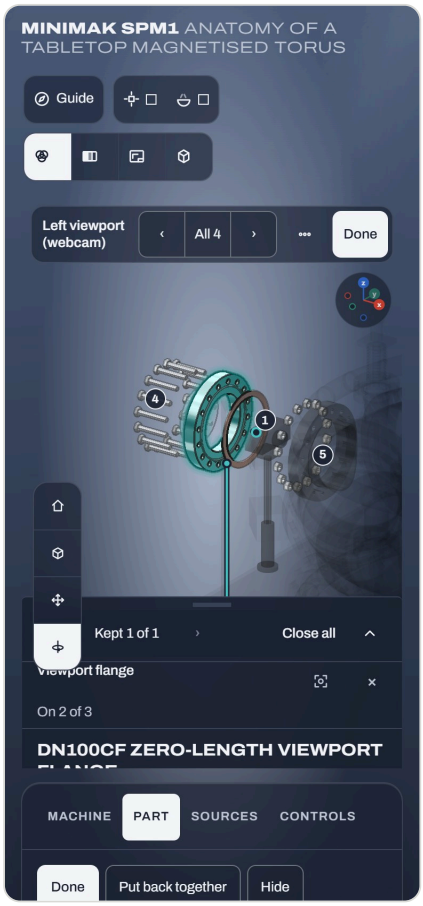
> **Zoomed out,** the lines inside each part fade and thin, as an illustrator draws a figure seen from further off; zoom in and every line comes back. A selected part always keeps all its lines.

13 Phones and tablets

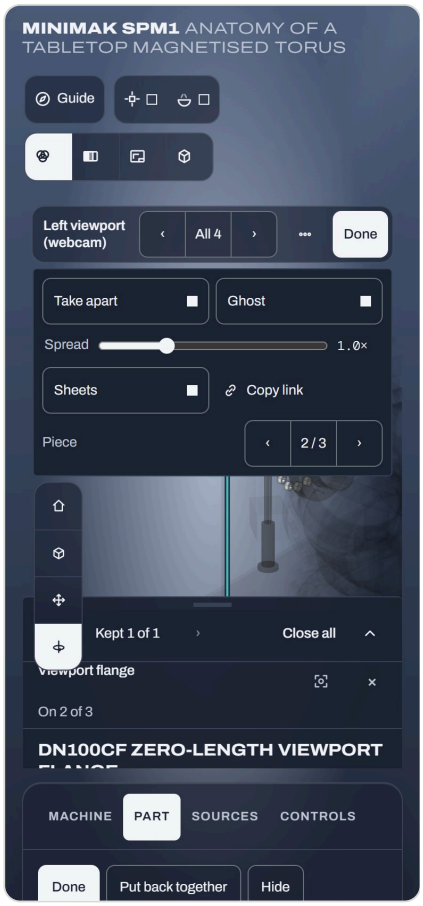
On a narrow screen the side panels stack under the model: scroll the page to read them. One finger turns the model, two fingers pan, and a pinch zooms.



The home view on a phone.



Inspecting: the bar is one row, and the sheets are a bottom sheet.



··· opens the rest of the inspect bar.

Inspecting on a phone

- **Tap** to select, **double-tap** to inspect.
- The inspect bar is one row: the part's name, one way to step, ··· and **Done**. ··· holds Take apart, Spread, Ghost, Sheets, Copy link and the pieces.
- The sheets are a **bottom sheet** that peeks under the part. Pull it up by its head (or tap the head) to read; push it down to peek again. Its arrows step through the kept sheets.

Hitting small parts

- A tap picks the smallest part within 24 px of your finger, so a tap on a flange beside a nut gets the nut.
- Inspecting, a tap **previews** a sub-part's sheet; **Keep** keeps it.
- **Loupe:** hold a finger still for half a second. A magnifier shows the view under it at 3×, with the name of what a tap there would pick. Slide to move it; lift to tap there.
- Every button is at least 44 px on a touch screen.

14 Links, settings and access

Share exactly what you see

The address in the browser follows what you are looking at: the part and piece, the kept sheets, the step, Show me, the view and the theme. Copy it, or press **Copy link** in the inspect bar, and anyone who opens it sees the same thing, even in a fresh browser. The browser's Back button returns to the part before.

`#part=gaskets&piece=right-back`	The right–back gasket, inspected
`&sheets=ofhc-copper-gasket`	…with its sheet kept
`&joint=1&step=2`	…with the joint opened at step 2
`#quiz=gauge,viewport,laser,needle,probe`	A quiz round of those five parts
`#theme=blueprint`	Opens in the Blueprint theme
`#desk=1`	Opens the control desk

A piece or sheet can be named by the start of its name, as `piece=right-back` above.

Settings

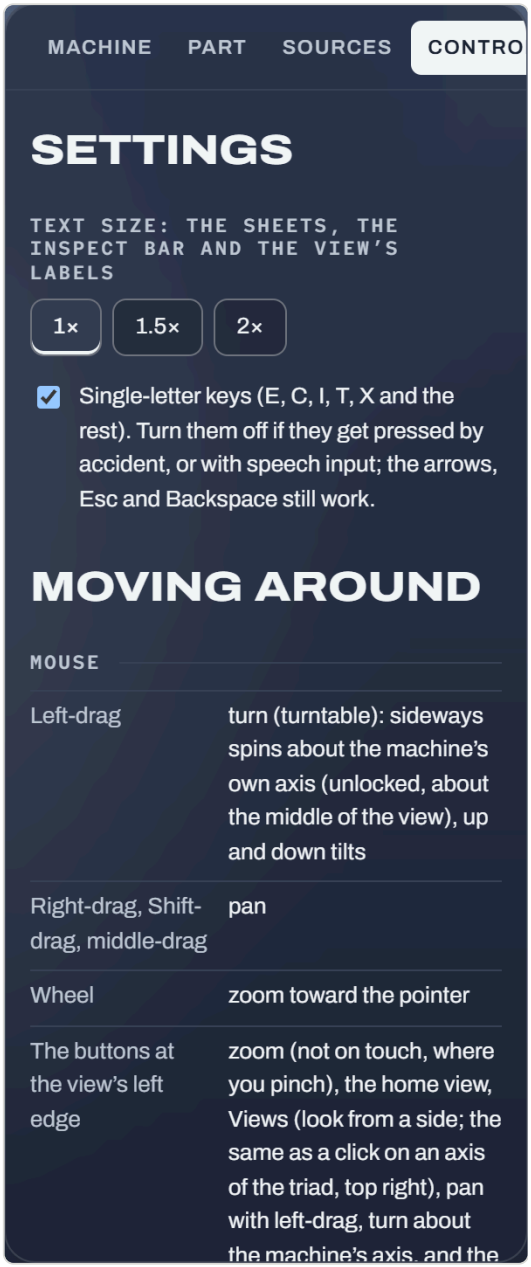
The **Controls** tab starts with two settings, which the page remembers:

- **Text size** 1×, 1.5× or 2× scales the sheets, the inspect bar, the view's labels and the tooltips, for a projector or a lecture hall.
- **Single-letter keys** can be turned off, for speech input or if keys get pressed by accident. The arrows, Esc and Backspace still work.

Below them, the Controls tab lists every mouse, touch and key binding.

Keyboard and screen readers

- The first Tab stop, **Skip to the assembly**, jumps to the list. In it, ↑ ↓ move, → ← open and close a group, Enter selects, Space hides or shows, and I inspects.
- The theme buttons are one Tab stop: ← → switch theme.
- On the groups' wheel, Enter opens a group, the arrows step through its parts, and Esc folds it.
- A status line announces what changes, and the 3D view has a text description of what it shows and from which side.
- With reduced motion turned on in your system, glides become cuts and the animation stands still.



The Controls tab: the settings, then every binding.

15 Keys and gestures

Click the model first so it has the keys. With single-letter keys turned off, only the arrows, Esc and Backspace work.

The view

← → ↑ ↓	Turn
Shift + arrows	Pan
+ −	Zoom
0	Home view (inspecting, the part)
A	Fit everything shown
1–4	Three-quarter, front, right, above
O / P	Left-drag turns / pans
R	Turn about the machine's axis, or the middle of the view
F	Fold or unfold the side panels

The machine

/	Find a part
E	Explode
[ ]	Less or more spread
C	Cutaway
H	Hide the selected part
I	Inspect the selected part

Mouse

Drag	Turn
Right-drag	Pan (also Shift-drag, middle-drag)
Wheel	Zoom toward the pointer
Click	Select; inspecting, keep a sheet
Double-click	Inspect; inspecting, Show me
Rest on a sub-part	Preview its sheet
Double-click space	Home view
Turntable ‹ ›	An eighth of a turn
Rest on the wheel	Fan a group's parts out
Click the emergency stop	At the control desk: trip it

Inspecting

T	Take apart
J	Open or close the joint
, .	Previous or next piece
G	Ghost on or off
S	Sheets on or off
Z	Show me
L	Floating sheets back into the column
Backspace	Back to the part you came from
Esc	One step back

Touch

One finger	Turn
Two fingers	Pan; pinch to zoom
Tap	Select; inspecting, preview a sheet
Double-tap	Inspect
Hold still	Loupe at 3×; lift to tap there

In the assembly list

↑ ↓	Move between rows
→ ←	Open or close a group
Enter	Select the part
Space	Hide or show it